

5. Write a C program to simulate:

a) Bankers' algorithm for the purpose of deadlock avoidance.

CODE:

```
#include <stdio.h>

int main() {
    int n,m;
    printf("Enter no. of rows and columns:");
    scanf("%d %d",&n,&m);
    int i,j,k;
    int alloc[n][m], max[n][m];
    printf("Enter the allocation matrix:\n");
    for (i=0;i<n;i++)
        for(j=0;j<m;j++)
            scanf("%d",&alloc[i][j]);
    printf("Enter the max matrix:\n");
    for (i=0;i<n;i++){
        for(j=0;j<m;j++){
            scanf("%d",&max[i][j]);
        }
    }
    int avail[m];
    printf("Enter available matrix:\n");
    for (i=0;i<m;i++)
        scanf("%d", &avail[i]);
    int need[n][m];
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)
            need[i][j] = max[i][j] - alloc[i][j];
    printf("NEED MATRIX:\n");
    for(i = 0; i < n; i++) {
        printf("P%d : ", i);
        for(j = 0; j < m; j++)
            printf("%d ", need[i][j]);
        printf("\n");
    }
    int finish[5] = {0};
    int safeSeq[5];
    int count = 0;
    while(count < n) {
        int found = 0;
        for(i = 0; i < n; i++) {
            if(finish[i] == 0) {
                for(j = 0; j < m; j++) {
```

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        if(need[i][j] > avail[j]) break;
    }
    if(j == m) {
        for(k = 0; k < m; k++)
            avail[k] += alloc[i][k];
        safeSeq[count++] = i;
        finish[i] = 1;
        found = 1;
    }
}
}
if(found == 0) {
    printf("\nSystem is NOT in safe state\n");
    return 0;
}
}
printf("\nSystem is in SAFE state\n");
printf("\nSafe Sequence:\n");
for(i = 0; i < n; i++) {
    printf("P%d", safeSeq[i]);
    if(i != n - 1)
        printf(" -> ");
}
printf("\n");
return 0;
}

```

OUTPUT:

```
Enter no. of rows and columns:5 3
Enter the allocation matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2
Enter the max matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3
Enter available matrix:
3 3 2
NEED MATRIX:
P0 : 7 4 3
P1 : 1 2 2
P2 : 6 0 0
P3 : 0 1 1
P4 : 4 3 1

System is in SAFE state

Safe Sequence:
P1 -> P3 -> P4 -> P0 -> P2

Process returned 0 (0x0)   execution time : 62.437 s
Press any key to continue.
|
```